

Emergence of augmented reality

The last 50 years have seen tremendous developments and advancements in the area of AR, both in terms of technologies as well as applications.

1962: Morton Heilig, a cinematographer designs a motorcycle simulator called Sensorama

1968: Ivan Sutherland creates AR system called The Sword of Damocles

1975: Myron Krueger created Videoplace, the original pioneer of VR and interactive art called AR System

1992: Tom Caudell and David Mizell of Boeing's Computer Services Adaptive Neural Systems Research and Development Project coin the term Augmented Reality

1994: Julie Martin creates the first AR theatre production called Dancing in Cyberspace

2001: Reitmayr and Schmalstieg create a mobile, multi-user AR system

2004: Mathias Mohring develops the first system for tracking 3D markers on mobile phones

2006: Nokia initiates Mobile Augmented Reality Applications (MARA) project

2008: Mobilizy launches Wikitude World Browser with AR

2009: ARToolkit brings AR to Web browsers

2013: Volkswagen MARTA app provides detailed information for service technicians. Audi AR app uses iPhone camera to provide vehicle features to end users

2016: Various VR and AR headsets are launched till date by IT and electronics tech giants like Samsung, LG, HTC, Microsoft and Facebook

can see exactly where the medical instruments are going—just like having X-ray vision. The system shows a 3D picture of the patient's inner body.

EyeDecide AR app. This app is created by OrcaMD, and it provides patient-engagement tools for health-care providers. Using EyeDecide, developers use the camera to simulate the vision of a patient affected by a certain condition. Basic features of this app are as follows:

1. Leverage interactive anatomical models to enhance patient understanding
2. Effectively educate patients by comparing normal and abnormal images and using 3D animation videos of most common conditions
3. Improve patient and surgical retention by helping patients make more informed decisions about the care

SimX. This is regarded as the professional-grade AR medical simulation software that replaces physical simulation with a customizable, high-definition, 3D virtual patient that can be projected onto any empty hospital bed. Whether the patient is obese, pregnant,

young, old, vomiting, missing limbs, bleeding or expressing any number of physical signs or symptoms, SimX software allows for reproduction of

patient's presentation with unprecedented visual fidelity.

The software builds complex cases in minutes using its powerful visual case building system, drags and drops events onto the field, determines the environment, sets up patient data, uses SimX's powerful case-monitoring and feedback system to see the case from each trainee's perspective, and adjusts case parameters on-the-fly.

Technology of medical simulation continues to rapidly evolve with a new announcement from immersive wearable headset manufacturer Meta that says, their new Meta 1 AR glasses have started being used in hospitals.

Key features of Meta 1 AR device are:

True-scale holograms. Allows you to see the physical and holographic worlds merge through 3D stereoscopic display in real size, depth and parallax

Marker-less surface tracking. Lets you look around the room and watch as holograms stay anchored to physical tables, floors and walls, thanks to low-latency, 360° tracking

Natural user interface. Allows you to grab, pinch and touch 3D objects in the real world, and drive a touch based holographic user interface

With SimX on Meta 1, you can practice using real instruments on virtual patients of any gender, age or having any medical issue. You can simulate with live action and in real time. This could include any clinical situation such as birth, heart attack or stroke.

AccuVein. AccuVein digitally maps and



Fig. 2: EyeDecide AR app



Fig. 3: SimX offers AR medical simulations through Meta 1 AR device